Nowcasting Israel's Quarterly GDP Growth

Introduction

Policymakers require timely and accurate assessments of economic activity to make informed decisions. In practice, however, official Gross Domestic Product (GDP) figures are released with significant lags, often several weeks after a quarter's end and are subject to revisions. This delay poses challenges for real-time decision-making in fast-moving economies. Nowcasting has emerged as a solution: using high-frequency data and statistical models to estimate the current state of the economy before official figures are available. Central banks in many countries have developed nowcasting models to guide monetary policy decisions, notably the Federal Reserve Bank of New York and the Bank of Israel among others (O'Keeffe and Petrova, 2025; Ginker and Suhoy, 2022). These models integrate a wide array of indicators to produce short-term forecasts (or "nowcasts") of GDP growth, enabling policymakers to respond to economic developments as they unfold rather than waiting for official data.

This paper presents a nowcasting model for estimating the quarterly GDP growth rate of Israel in real time, with a particular focus on the needs of fiscal policymakers. The model, designed for the Israeli Ministry of Finance, draws on methodologies pioneered by central banks while emphasizing interpretability and policy relevance. Beyond generating forecasts, it provides insight into the drivers of GDP fluctuations, allowing officials to identify which sectors or factors are contributing most to growth at any given time. By producing estimates ahead of the official GDP release, the model supports more timely and proactive fiscal policy responses. In doing so, it aims to bridge a gap in policy tools: while nowcasting is common in monetary policy circles, analogous tools for fiscal authorities are less developed. This study's contributions are twofold. First, it adapts the well-known Taylor Rule framework, originally a monetary policy rule, to highlight the importance of real-time output information for both monetary and fiscal policy in a small open economy context. Second, it develops a dynamic factor model (DFM) nowcasting framework tailored to Israel's economy, where users can

select and modify the set of input indicators and gain insights into how each indicator contributes to the latent factors that capture their shared dynamics over time.

The remainder of the paper is organized as follows: Section 2 provides the theoretical background on the Taylor Rule and its evolution, including applications in small open economies and potential fiscal analogs. Section 3 discusses the motivation for real-time GDP estimation for fiscal policy. Section 4 outlines the nowcasting model's methodology, including the data structure, factor blocks, estimation approach, and interpretability features. Section 5 concludes with remarks on policy implications and future extensions.

Theoretical Background: The Taylor Rule and Policy Response in Real Time

John Taylor's seminal work (Taylor, 1993) introduced a simple rule-of-thumb for monetary policy that has since become a benchmark in macroeconomic policy discussions (Brookings Institution, 2015). Taylor posited that the central bank's policy interest rate (e.g. the federal funds rate in the U.S.) should respond systematically to two key deviations: (1) inflation from its target, and (2) output (GDP) from its potential (the output gap). In its original form, the Taylor Rule can be summarized as:

- *Policy Rate* = *Neutral Real Rate* + *Current Inflation* + 1.5 × *(Inflation - Inflation Target)* + 0.5 × *(Output Gap)*.

This rule implies that for each percentage point that inflation exceeds its target, the central bank should raise the nominal interest rate by 1.5 percentage points, and for each percentage point that output exceeds potential, it should raise the rate by 0.5 percentage points (Brookings Institution, 2015). Likewise, if inflation is below target or output is below potential, the rule prescribes lowering the interest rate. The Taylor Rule captured the U.S. Federal Reserve's behavior in the late 1980s and early 1990s quite well, and it demonstrated the value of a predictable, transparent policy rule. When inflation is at target and output is at potential (i.e. zero output gap), the rule yields a policy rate equal to the long-run neutral rate (which Taylor originally assumed to be around 2% in real terms) (Brookings Institution,

2015). By construction, this approach balances price stability and output stabilization in setting monetary conditions.

Evolution and Extensions: Over the past decades, the Taylor Rule has been extended and modified in various ways. Researchers and practitioners have experimented with different coefficients and additional variables to better fit historical policy or to derive optimal rules. For instance, some "balanced-approach" variants place a larger weight on the output gap (coefficient of 1.0 instead of 0.5) to reflect a greater emphasis on output stabilization (Goldberg, 2024, p. 269). Others introduce interest rate smoothing (adding inertia so that the policy rate adjusts gradually) or make the rule forward-looking (responding to expected future inflation and output rather than current values). Crucially, in the context of small open economies like Israel, simple Taylor-type rules often require augmentation to account for external influences. Open economy macroeconomic theory suggests that domestic interest rates cannot be set in isolation from the rest of the world. Capital mobility and exchange rate dynamics mean that foreign conditions, such as the global interest rate environment or terms of trade influence domestic inflation and output. Thus, policymakers may implicitly or explicitly respond to such external factors. Caputo and Herrera (2013) derive an efficient Taylor Rule for a small open economy, showing that the policy rate should adjust not only to domestic inflation but also in line with foreign interest rates and other external shocks to maintain optimal outcomes. In their New Keynesian model, the *neutral interest rate* in a small open economy is directly linked to the world interest rate and domestic productivity, ensuring that monetary policy reacts appropriately to both domestic and foreign developments. Empirically, they find that inflation-targeting central banks in open economies do tend to respond to movements in foreign interest rates (such as the U.S. federal funds rate) in addition to reacting to domestic inflation and the output gap. This aligns with the intuitive observation that countries like Israel, which are open to trade and capital flows, cannot ignore global financial conditions when setting domestic policy.

The Taylor Rule's influence has largely been in the monetary policy sphere, but the underlying principle, systematic countercyclical response is applicable more broadly. In the fiscal policy realm, there is no single analogous rule as universally recognized as Taylor's rule for monetary policy. However, governments often implement fiscal rules or guidelines

that aim for sustainability while allowing for countercyclical action. For example, some countries target a structural budget balance, which requires estimating the output gap in real time to distinguish cyclical vs. structural deficits. This means fiscal authorities attempt to tighten policy (reduce deficits) when the economy is running above potential and loosen policy (allow larger deficits or stimulus) when the economy is below potential, in order to stabilize output. Such behavior can be seen as a fiscal analog to the Taylor principle: responding to economic conditions in a rule-based manner. In practice, automatic stabilizers (like progressive taxes and unemployment benefits) already provide an automatic fiscal response to output fluctuations, and policymakers may add discretionary measures on top of that during recessions or booms.

The need for real-time information: Whether for monetary or fiscal policy, a critical assumption behind any rule-based response (including Taylor's rule) is the availability of timely and accurate information on the state of the economy. Taylor (1993) implicitly assumed that policymakers know the current inflation rate and output gap when setting interest rates. In reality, measuring the output gap in real time is challenging; it requires knowing current GDP (which is only published with a delay and often revised) and potential GDP (which is unobservable and must be estimated). Data lags and revisions can lead to policy missteps if the real-time assessment is off. Output gap estimates can differ markedly from later revised data, which has implications for policy rules and their performance. This underscores the importance of developing tools to gauge the economy's current state as accurately as possible in real time. For small open economies, additional complexity arises from needing real-time indicators of external conditions (e.g. global demand, commodity prices, exchange rates) since these can shift rapidly and feed into domestic output and inflation.

In summary, theoretical and empirical developments around the Taylor Rule highlight two key points for this study: (1) Policy effectiveness (monetary or fiscal) depends on real-time economic assessment, and (2) Small open economies like Israel have to broaden the information set to include external variables. These insights motivate the nowcasting approach we take. By using high-frequency indicators, a nowcasting model can provide the timely estimates of output growth needed to inform a Taylor-type policy response – for the

central bank and, by analogy, for fiscal authorities. The next section turns to the specific motivation for real-time GDP estimation from the perspective of the fiscal authority (Ministry of Finance), building on this theoretical backdrop.

Motivation for Real-Time GDP Estimation in Fiscal Policy

Fiscal policymakers share with central bankers the need for up-to-date information on economic conditions, but their decisions involve tools like government spending, taxation, and borrowing rather than interest rates. The Ministry of Finance (MoF) in Israel plays a key role in short-term stabilization and long-term planning, and timely GDP estimates are crucial for both. Unlike a central bank that meets regularly to set interest rates, fiscal authorities operate through budgets and policy packages that often have longer lead times, yet they still must decide on stimulus or austerity in response to economic developments. Having a real-time read on GDP growth allows the MoF to calibrate fiscal responses (such as emergency spending, tax relief measures, or conversely, fiscal tightening) before official data confirm a downturn or overheating. This proactive stance can improve economic outcomes by addressing issues early, rather than reacting with delay when official statistics arrive.

There are several reasons why real-time GDP nowcasting is particularly valuable for fiscal management in a small open economy like Israel. First, Israel's economy is influenced by a wide range of factors; from high-frequency domestic indicators (e.g. monthly tax receipts) to international developments. Shocks can come from sudden changes in the global environment (such as a spike in oil prices, financial market turbulence, or geopolitical events) and these can quickly ripple through trade, exchange rates, and investor confidence. Real-time monitoring helps distinguish whether a mid-quarter slowdown in growth is being driven by, say, a drop in export orders versus a domestic demand issue, which in turn informs the appropriate fiscal response. As noted earlier, external shocks can rapidly alter domestic output and inflation in Israel. For example, a sharp depreciation of the shekel might boost inflation and affect GDP through trade, requiring a different fiscal consideration compared to a scenario where domestic consumer demand is faltering.

Second, Israel's fiscal framework includes elements that depend on current economic conditions. Tax revenues are highly sensitive to the business cycle, in expansions, revenues

surge (sometimes creating pressure to cut taxes or increase spending), while in recessions revenues slump (just as social spending needs rise). The Ministry of Finance monitors monthly tax collection data closely as an indicator of economic activity. A nowcasting model that incorporates tax revenue data can provide an early read on GDP by capturing these trends. This tax data is typically released within two weeks after month-end and offers timely insight into current economic activity, broken down by sector, making it particularly well-suited for nowcasting. Similarly, real-time tracking of indexes following industrial production, salary and employment and the consumer trust can signal economic turning points. By integrating such information, nowcasts enable the MoF to adjust in-year budget execution; for example, accelerating planned investments or delaying certain expenditures to stabilize the economy.

Third, timely GDP estimates support debt and deficit management. If a downturn is detected early, the government can anticipate lower tax revenues and potentially higher expenditures, allowing it to revise deficit projections and financing plans sooner. Conversely, in an upswing, better-than-expected growth (detected via nowcasts) might create room for reducing borrowing or building fiscal buffers. In both cases, acting on real-time information can enhance fiscal credibility by avoiding large surprises in deficit outcomes. It also feeds into communication: the government can explain its mid-year budget adjustments or stimulus packages by pointing to real-time economic indicators and model-based nowcasts, lending analytical support to policy decisions.

Moreover, by developing its own nowcasting capacity, the Ministry of Finance is not fully dependent on central bank assessments, which may reflect different mandates or indicator priorities. While the Bank of Israel's models are primarily geared toward monetary surveillance, a dedicated fiscal model can prioritize variables such as tax receipts, government outlays, or sector-specific activity relevant to budget execution. This independence allows the Ministry to tailor the data inputs and analytical focus to its own needs—be it intra-year budget planning, revenue forecasting, or identifying early shifts in economic behavior that may not yet trigger a monetary response. In doing so, nowcasting becomes not just a forecasting tool, but a strategic instrument for real-time fiscal

governance, offering insights for targeted interventions, sectoral diagnostics, and political decision-making.

In summary, the motivation for real-time GDP nowcasting in fiscal policy is to enable proactive and well-informed fiscal management. By having an early estimate of quarterly growth, the Ministry of Finance can make data-driven decisions on fiscal stimulus or consolidation, adjust policies to cushion against external shocks, and coordinate effectively with monetary policy. The next section describes the methodological framework of the nowcasting model developed to meet these needs.

Methodology: A Dynamic Factor Model for GDP Nowcasting

Schematic Workflow: Raw Data → Real Prices → Seasonal Adjustment → Transformation to Stationary → DFM → Forecast

Model Framework and Mixed-Frequency Structure

The proposed model was developed for the Israeli Ministry of Finance as an additional tool in the Ministry's forecasting toolbox, complementing other methods used to predict changes in quarterly GDP. Developing such a model involves a range of considerations, including the nature of the input data, the target variable, the publication schedule of each series, the intended use of the model, and the ability to interpret its results. There are inherent trade-offs in these design choices: emphasizing one aspect of the model (for example, ensuring high interpretability of the results) often comes at the expense of employing certain advanced statistical techniques. Conversely, choosing methods that allow the model to handle limited data availability or a large number of predictors can detract from the interpretability of the outcomes. The model design therefore reflects a balance among these factors to meet the Ministry's needs.

Objectives

The nowcasting model is designed to meet several key requirements for the Ministry of Finance:

- Real-Time Forecasting: As implied by the term *nowcasting*, the model must produce forecasts in real time. This requirement influences the choice of data (favoring

indicators available with minimal delay), the handling of publication lags and synchronization of data releases, and the statistical methods employed. In particular, the model needs to accommodate missing or partially available data in real time, whether through methods of imputing missing observations or by selecting a model structure that can handle asynchronous data due to different release schedules or frequencies.

- Utilization of a Large Data Panel: The model should be able to ingest a broad panel of indicators (on the order of 50 monthly series) and make use of as much relevant information as possible. To manage this, the framework includes a filtering mechanism to exclude variables whose contribution is not significant and that might introduce excessive "noise" into the relationships between the predictors and the target variable. This ensures the model focuses on the most informative indicators without being diluted by irrelevant data.
- Result Interpretability: The final requirement is that the model's outcomes should be interpretable. It should be possible to understand which variables or factors are driving the nowcast and to clearly present the model outputs. This emphasis on interpretability guides the modeling approach, so that in addition to producing an accurate point forecast, the model provides insight into the underlying drivers of GDP growth as estimated in real time.

Data

A methodology introduced by the Bank of Israel (Ginker and Suhoy, 2022) involved the use of a monthly proxy to estimate quarterly GDP in real time. Following this principle, and after adjusting for differences in publication lags, I searched for suitable proxy variables. Employment indicators, such as the employment rate and the number of salaried jobs, exhibited strong correlations with quarterly GDP, similar levels of correlation observed among "The Combined Index of BoI". However, both posed significant limitations: they are published with a long delay relative to the end of the reference month, and their historical availability is limited, beginning over 15 years later than most other indicators in the dataset. To overcome these limitations, I constructed a broader dataset composed of variables that

are indirectly related to the original proxies, but are published earlier and more consistently. For example, since salaried employment is strongly correlated with GDP yet released with a significant delay, I identified tax-based indicators, such as income tax receipts, that reflect similar economic dynamics and are available much sooner. This strategy allows the model to capture the relevant economic signals while maintaining real-time applicability.

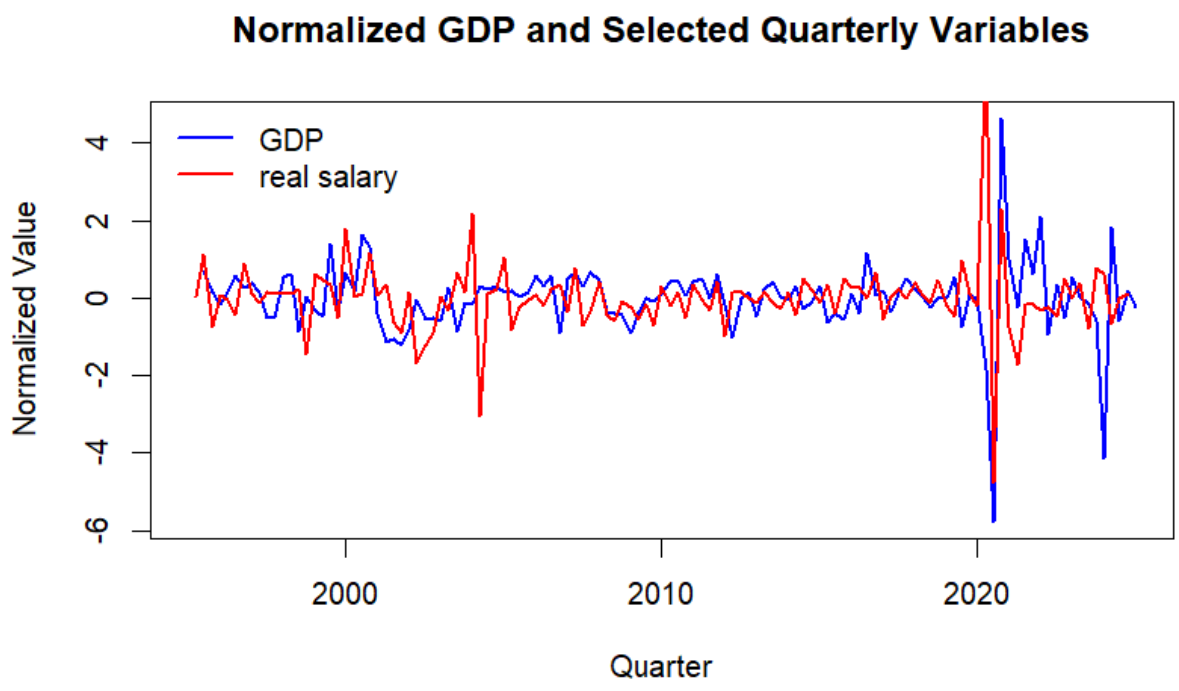

Figure 1: correlation between lagged "real salary" and quarterly change in GDP

The dataset consists of roughly 50 monthly indicators covering various aspects of the economy. For clarity, these series are grouped into thematic categories, similar to the "block" structure used in the New York Fed's nowcasting model (Almuzara et al., 2023). The main data groups are:

- Personal Income Taxes: This group comprises about 12 monthly variables related to wage and self-employed income. These tax data are published very quickly, usually about two weeks after the end of the month, allowing almost immediate use once the observation period concludes. Figure 1 (noted above) shows the strong relationship between salaried employment and GDP. However, because the official salaried

employment figures are released with a considerable delay compared to the tax data, the personal income tax receipts serve as a timely proxy indicator to help predict the change in quarterly GDP.

- Corporate Taxes: The corporate tax group serves as an indicator of business sector activity in Israel. It captures changes in corporate revenues and expenditures, distinguishing between revenues from sales and those related to financial activities of firms. Similar to personal taxes, corporate tax data are typically available about two weeks after month-end, much earlier than conventional real-sector indicators like the industrial production index or other business activity indices. This timeliness makes corporate taxes a useful early signal of business conditions.
- Consumption (Indirect Taxes): This set of indicators is directly related to a component of GDP, namely private consumption (denoted as C in national accounts). It consists of measures such as domestic VAT receipts, fuel excise taxes, and other sales taxes. These consumption related tax revenues provide insight into household spending activity in near real time, since they reflect the volume of consumption expenditures subject to taxation.
- Import Taxes: This group of indicators ties directly to another GDP component, net exports (EX – IM). It includes measures like import duties and import VAT. High-frequency data on import taxes can serve as a proxy for import activity (and by extension, net export contributions to GDP), again available on a timelier basis than the official trade statistics.
- Real Estate: This category combines real-estate related tax data with real estate market indicators. It reflects the component of GDP related to investment in capital (denoted I, particularly residential investment). The variables include real estate transaction taxes (such as purchase taxes and capital gains taxes specific to property) and market activity indicators like the number of home sales or housing starts. Together, these capture the pace of activity in the real estate sector, which is an important driver of investment in the economy.

- Real Activity Indicators: This group contains several indicators that track different facets of overall real economic activity. Key examples are the Industrial Production Index, the Consumer Confidence Index, the Composite State-of-the-Economy Index (a monthly coincident index), and the Trade and Services Revenue Index. Although these indicators are published with a lag (often several weeks after the end of the month), they provide an informative snapshot of real economic activity within the quarter. They help the model gauge the current momentum of the economy from multiple angles.
- Labor Market: This group includes labor market indicators such as employment indices and additional data on unemployment and labor force participation. As noted earlier, there is a strong and stable correlation between the monthly salaried employment index and quarterly GDP, so these labor market series should reflect the monthly trajectory of output up to the end of the quarter. However, high-frequency unemployment and participation rate data are available only from Q2 2012 onward in monthly form, which created challenges in preserving these series for modeling. While these labor market indicators are intended to contribute information about the trend of GDP, their limited historical availability (and the resulting gaps in the data) restricts their use to a relatively short sample, potentially introducing certain biases.
- Financial Markets and Global Activity: Given that Israel is a small open economy exposed to external influences, this category accounts for indicators reflecting those external factors. It includes variables such as Israeli financial market indices (e.g. stock market indices) and global stock indices, as well as the real effective exchange rate (REER) of the shekel and the foreign exchange reserves of the Bank of Israel. These variables provide measures of external shocks or conditions, for instance, the REER captures the relative value of the shekel (affecting trade competitiveness), and reserves can indicate interventions or external balance conditions. Including such indicators ensures that global economic and financial developments are factored into the nowcast to the extent that they influence domestic output.

Transformations

In order to use these indicators in a forecasting model, known sources of noise in the raw data must be removed so that hidden underlying trends can be identified. Key data transformations applied were as follows:

1. Seasonal Adjustment: Recurring seasonal patterns were removed from the series. This was done using the Ministry of Finance's seasonal adjustment method, which is adapted to the Israeli calendar. Seasonal adjustment eliminates regular seasonal fluctuations that could obscure the true economic signal.
2. Stationarity (Detrending): Statistical tests for stationarity (KPSS and ADF tests) and visual inspections were conducted to determine whether each series contained a unit root or a persistent trend. Based on these diagnostics, deterministic trends were removed from the series as needed. Ensuring stationarity is important for the time series modeling, as it focuses the model on short-term fluctuations rather than long-term growth trends.
3. Inflation Adjustment: Nominal variables were converted to real terms by adjusting for price level changes. In practice, this transformation removes the effect of general inflation from the data. By deflating nominal series, the model isolates the real (volume) movements, preventing inflation driven trends from being mistaken as changes in real activity.
4. Currency Adjustments: For variables denominated in or affected by foreign currency (for example, measures that involve exchange rates or foreign currency values), appropriate adjustments were made to neutralize exchange rate effects. This puts all series on a comparable footing (in real local currency terms) and ensures that movements reflect real activity or relative price changes rather than currency fluctuations.

All these transformations were aimed at distilling the underlying signal that the model is trying to capture, the variation in quarterly GDP that is not explained by trivial or predictable movements like regular seasonal swings, long term growth trends, or general price inflation. The preprocessing pipeline was implemented as a flexible tool: a user can take any set of

input series and pass it through this pipeline to obtain transformed series that are suitable for statistical modeling. This design allows for flexibility and experimentation with different groups of variables, as the same transformations will prepare any chosen dataset for use in the model.

Publication Delays

Even with monthly indicators (which in principle should be suitable for early prediction of quarterly GDP), one must account for publication lags in these data. Each series is released with its own delay after the reference period, so the timing of data availability must be synchronized for real-time forecasting. In the constructed dataset, each monthly observation is recorded not at the time of the activity it represents, but at the time of its release. In other words, if a series measures the activity of a given month, we assign that observation to the calendar date when it became available to analysts, rather than the date of the activity itself. This adjustment enables an accurate simulation of real-time forecasting at any given point in history; one would only use the information that had actually been published up to that date.

Release Lag in days (from end of month)	***Push***
<30	*1 month forward*
>30	*2 months forward*
>60	*3 months forward*

Table 1: summarizes the typical publication lag for the monthly indicators and the corresponding shift applied to align them with availability.

Specifically, if an indicator is typically released less than 30 days after the end of its reference month, its value is treated as available one month after the reference month. If the release lag is more than 30 days (but within 60 days), the data is shifted two months forward, and if the lag exceeds 60 days, the data is shifted three months forward. By applying this rule, the

dataset is restructured such that an observation "appears" in the period when it would historically have been available. This ensures that when we conduct real-time nowcasting tests, we only use data that would have been accessible at that point in time. Furthermore, this structure allows the model to be used in live real-time: as new data are released each month, they can be incorporated immediately in the nowcast, with each series automatically aligned to the correct timeframe.

Dynamic Factor Model (DFM)

To address the Ministry's requirements, the model employs a Dynamic Factor Model, a popular framework particularly well suited for real-time forecasting with many variables at mixed frequencies. Indeed, variations of the DFM approach are used by numerous central banks and institutions for nowcasting economic activity. The model was implemented in R using the DFMS package, taking advantage of the Ministry's existing use of R for analytical work (ensuring compatibility and ease of use).

In essence, a Dynamic Factor Model (DFM) assumes that a large number of observed time series can be driven by a relatively small number of unobserved factors. Our model ingests a set of monthly indicators (the high-frequency data) and quarterly GDP (the low-frequency target) and extracts a few latent factors that capture the common co-movement of the many series at each point in time. These latent factors summarize the information in the broad dataset, and at the quarterly frequency they align with the movements in GDP, these can later be extracted and used as regressors in a bridge model. Such properties make the DFM ideal for this task: it can handle high-frequency information and effectively "fill in" the monthly evolution of a lower-frequency series (quarterly GDP) by leveraging the shared dynamics captured by the factors.

Tools

We developed auxiliary tools to assist in specifying and applying the DFM. These tools help determine important model parameters automatically and allow users to experiment with different input sets:

- Factor Number Selection: The toolkit includes a statistical test (ICr) (Bai and Ng, 2002) that suggests an optimal number of factors. This information criterion aims to

find a relatively small number of factors that still explains the majority of variation in the dataset. By selecting the number of factors in a data driven way, we ensure the model remains parsimonious without sacrificing significant information.

- Lag Order Selection: After choosing the number of factors, a second diagnostic is used to determine the appropriate lag order in the factor dynamics (i.e. how many past values of the factors to include in the state-space model). This is essentially selecting the persistence of the latent factors. The decision in our implementation is guided by visual inspection of the criterion's output curve (a form of scree plot for lag lengths). In our case, the plot exhibited a clear "elbow" around p = 2-5, suggesting that two lags for the factor autoregression provide a good balance (capturing the needed dynamics without overfitting).

These tools make the model configuration process more systematic and robust, allowing quick recalibration if the set of input variables is changed or updated. A user can input a different panel of indicators and easily obtain recommendations for how many factors to extract and how many lags to include, streamlining the model re-specification for new datasets.

EM Algorithm and Data Completion

The DFM estimation employs an extended Expectation-Maximization (EM) algorithm to handle missing data, as real-time data are by nature incomplete (with staggered releases and mixed frequencies). We use an extension of the standard EM algorithm known as the "BM" method (Bańbura and Modugno, 2010). The essence of this algorithm is to iteratively fill in missing observations and estimate the model parameters until convergence. It addresses both types of missing data inherent in our problem: "ragged edge" missingness due to different publication timings (monthly vs. quarterly data, and variations among monthly series) and any internal gaps within a series.

In each iteration, given an estimate of the factors and model parameters, the algorithm imputes the missing values using the estimated relationships between the observed variables and the latent factors. This means it can perform "horizontal" completion (filling in missing values for a given time period across different series) as well as "vertical" forecasting

(predicting ahead for periods with no available data yet, effectively nowcasting future values). After filling in the data, the algorithm uses a Kalman smoother to re-estimate the latent factors given the completed dataset, refining the factors and the imputed values. This iterative process of expectation (filling missing data) and maximization (re-estimating factors and parameters) continues until the solution converges. The outcome is an estimate of the latent factors and a completed data matrix (with nowcasts for missing entries) that are most consistent with each other and with the observed data, given the model structure.

Forecasts

Using the above model, we consider two approaches to produce nowcasts for quarterly GDP in real time: an internal model-based forecast and an external bridging approach.

Internal Approach

The first approach generates a direct nowcast from within the DFM using the built-in EM algorithm described earlier. In practice, we take all data available up to a certain point within a quarter (for example, through the end of February for forecasting Q1) and let the model forecast the missing observations up to the quarter's end. This involves a 1 or 2 steps ahead prediction in the state-space, effectively asking the model to fill in the not-yet-published monthly data (and the quarterly GDP) for the remainder of the quarter. For instance, from a vantage point in mid-Q1 (end of February), the model can project ahead two months to April 1st and output a nowcasted value for Q1 GDP. This internal nowcasting method yielded good results in our evaluation relative to a baseline model. Notably, as more information is incorporated within the quarter, the nowcast converges closer to the eventual actual GDP, and the prediction errors decline. An advantage of this approach is that it is easy to produce (fully automated within the model) and it delivers the target forecast directly. However, a key drawback is that the procedure functions as a "black box." The complex state-space estimation means that back-engineering the exact equation or understanding the precise contribution of each variable in the forecast is very difficult. The internal DFM forecast cannot be readily expressed in a simple interpretable form, nor easily aligned to traditional macroeconomic equations, which limits the transparency of this approach.

Bridge Approach

The second approach, often called a bridging model, also fulfills the real-time forecasting requirement and handles missing mid-quarter data, but in a more transparent way. In this approach, the latent factors estimated by the DFM are used as regressors in an external ordinary regression model for GDP. We can use the factors either at the end of the quarter (for a contemporaneous nowcast, denoted h = 0) or use factor values projected ahead (h = 1, 2, 3 months ahead within the DFM) as inputs to the regression. This means that even mid-quarter, we can generate factors (including forecasts for the remaining months of the quarter) and plug them into a separate equation to predict quarterly GDP. Unlike the internal approach, the bridge method is much simpler and more intuitive to analyze. The DFM produces factors along with a factor loading matrix that describes each variable's contribution to each factor. With an external regression that provides coefficients on those factors for predicting GDP, we can dissect the results: we are able to infer which groups of variables (through their factor loadings) are most influential in driving the forecast. In other words, the combination of the factor loading information and the regression coefficients allows us to trace forecast contributions back to the original indicators, greatly improving interpretability. This approach yields a more straightforward explanation of the nowcast, at the cost of introducing an additional modeling step.

Results

The model's real-time nowcasting performance was evaluated against a baseline benchmark (GDP Random Walk). The internal DFM nowcasts showed a very significant improvement in accuracy relative to the baseline model. Among the different forecast horizons within each quarter, the nowcast made two months before the quarter's end (e.g., using information up to the end of February to predict Q1) achieved the highest accuracy (lowest RMSE). The nowcast made one month before quarter's end was the next best, followed by the nowcast at the very start of the quarter (three months ahead of completion), and finally the nowcast on the official release date. Notably, by the release date of the GDP (after the quarter is over), all monthly data for that quarter are available, and at that point the DFM's direct nowcast can be further improved by leveraging the factors in a bridging regression. In essence, the bridge

model can add additional accuracy on the final nowcast when the quarter is complete, by calibrating how the factors translate into GDP with a simple regression.

Horizon	***RMSE/Baseline RMSE***	***MAE/Baseline MAE***	***Directional Accuracy***
1	*0.336*	*0.407*	*0.67*
2	*0.265*	*0.280*	*0.93*
3	*0.818*	*0.783*	*0.67*

Table 2: Results Analysis, errors compared to baseline model and directional accuracy in the test period (Q4 2020-Q4 2024)

The substantial reductions in errors at the 1–3 month-ahead horizons highlight the gains from using the DFM-based nowcasting approach. The relative improvement is largest for the mid-quarter forecasts (reflecting the model's ability to exploit partial information).

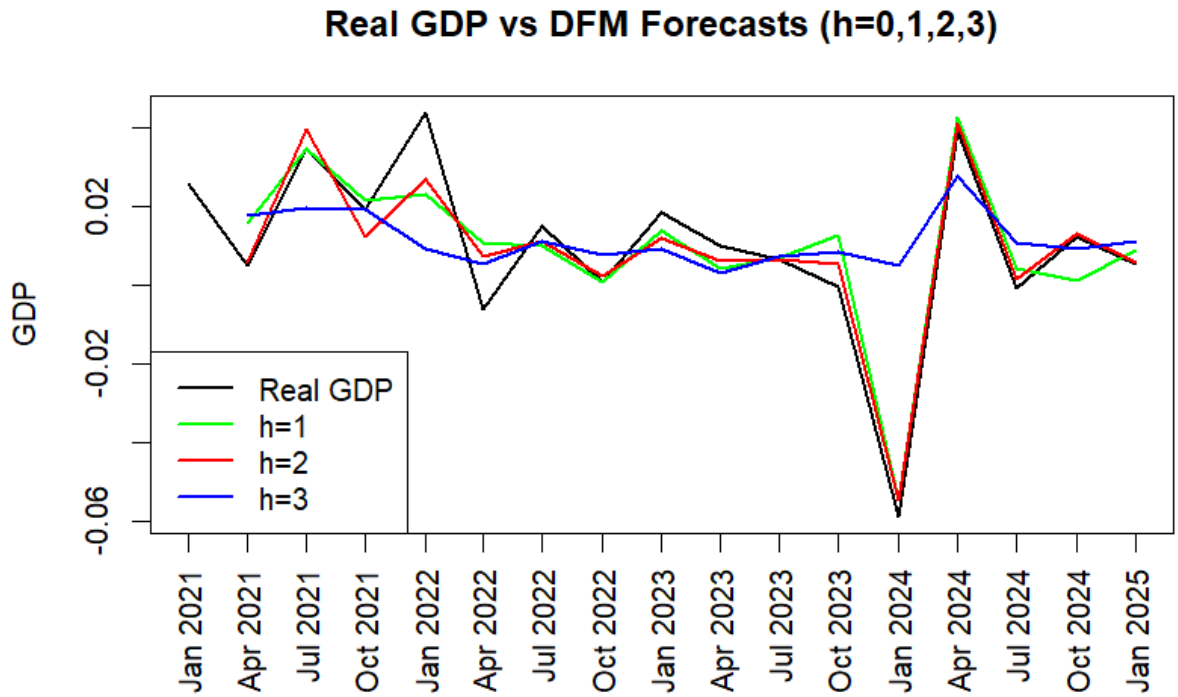

Figure 2: Pseudo nowcasting results on the test period with different forecast horizons using "Internal Approach"

The plotted nowcasts illustrates how the model tracks the evolving actual GDP. As expected, as the quarter progresses and more monthly information is incorporated, the nowcasts converges toward the actual realized GDP.

Lastly, examining the results at h = 0 (the end-of-quarter release date), the internal approach did not yield a substantial improvement over the baseline model. However, using the extracted factors as regressors in an external bridge model did lead to noticeably better forecast accuracy.

Model (at h=0)	*RMSE/Baseline RMSE*	*MAE/Baseline MAE*
DFM	*0.965*	*1.002*
Bridge	*0.802*	*0.849*

Table 3: Evaluating and comparing nowcast results at end of quarter day for internal DFM forecast and Bridge model using factors

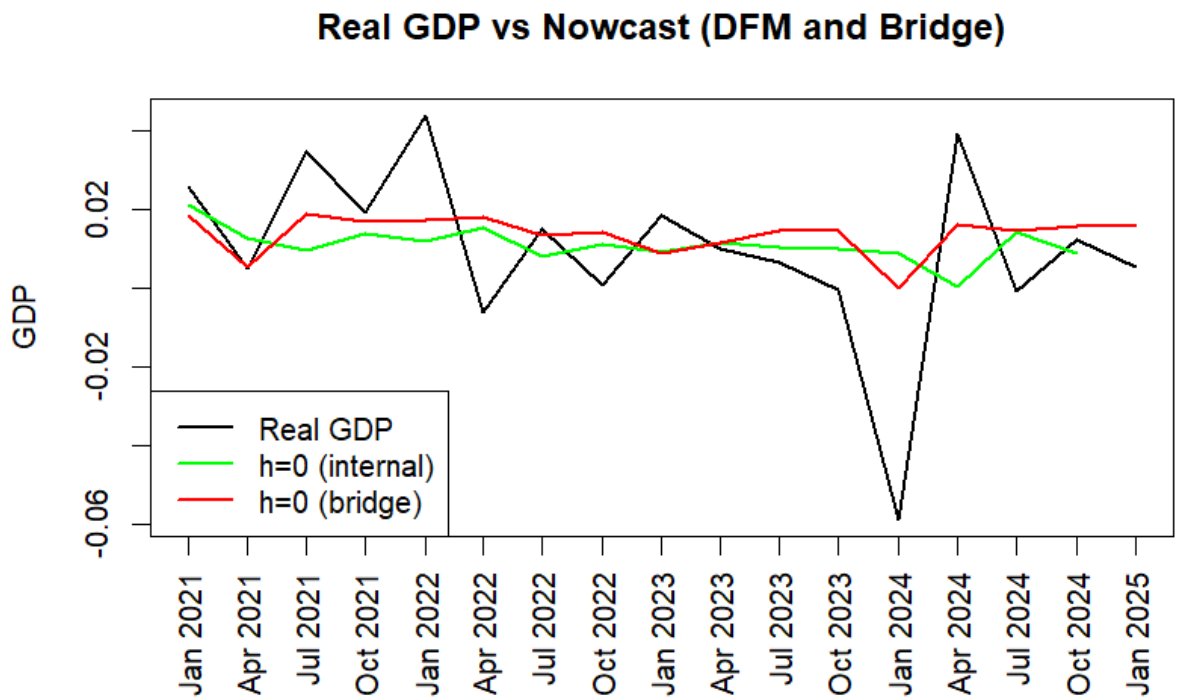

Figure 3: Comparison of pseudo-nowcast results at h = 0 shows that the bridge approach produced lower forecast errors than the internal DFM method.

The graph above illustrates a potential improvement to the model. While the forecasts produced by the internal DFM approach on the GDP release date did not significantly outperform the baseline model, using the factors within a bridge model rather than relying on the internal method, did yield improved forecast accuracy at the time of release (h = 0).

Summary

At the outset of this study, we discussed the theoretical motivation for real-time nowcasting of macroeconomic data and GDP in particular, including the importance of GDP in the Taylor Rule framework and the implied need for timely output information in policy response. From that motivation, the Ministry of Finance's need to monitor economic activity in real time was established, along with the unique requirements and adjustments necessary for a nowcasting tool that utilizes a broad dataset with ragged edges (staggered data availability) to provide up-to-date forecasts. Given these constraints and requirements, we presented the considerations behind the development of each component of the model and how each choice serves the Ministry's needs. Throughout, emphasis was placed on readying the model for real-world use, i.e. ensuring it can forecast with missing and delayed data by properly aligning release schedules, while also providing statistical justification for each component or transformation. The result is a stable model that generalizes well and is capable of delivering reliable nowcasts under a range of conditions.

The model demonstrated very strong results for nowcasting during the quarter, markedly improving forecast accuracy when data are incomplete. However, it does not outperform some of the Ministry's other forecasting tools at the end of the quarter, once all information has come in. This is to be expected, as when the quarter is complete and all data are available, simpler models can perform similarly well. Nonetheless, considering the full suite of tools available to the Ministry, the DFM-based nowcasting model is a valuable and useful addition. By providing a structured real-time estimate and combining its output with other tools and expert analysis, the nowcasting framework can increase the certainty and confidence of the Ministry's forecasts at various points in time before the official data is released.

Finally, the toolkit developed here also forms a basis for further extensions. Its modular structure allows analysts to adopt different nowcasting approaches, swap out components of the model, or retrain the model as needed. These features make the tool adaptable to changing requirements or the incorporation of new techniques in the evolving field of time series forecasting. As the data environment and methodologies continue to advance, this nowcasting framework can be updated and expanded, ensuring that it remains at the frontier of real-time economic monitoring for policy use.

References

Almuzara, Martin, Katie Baker, Hannah O'Keeffe, and Argia Sbordone. 2023. *"The New York Fed Staff Nowcast 2.0."* Federal Reserve Bank of New York Staff Reports, no. 1078, June.

Bai, Jushan, and Serena Ng. 2002. *"Determining the Number of Factors in Approximate Factor Models."* Econometrica 70 (1): 191–221.

Bańbura, Marta, and Michele Modugno. 2010. *"Maximum Likelihood Estimation of Factor Models on Data Sets with Arbitrary Pattern of Missing Data."* Working Paper Series no. 1189. European Central Bank.

Brookings Institution. n.d. *"The Taylor Rule: A Benchmark for Monetary Policy?"* Accessed April 2025. https://www.brookings.edu/articles/the-taylor-rule-a-benchmark-for-monetary-policy/.

Caputo, Rodrigo, and Luis Oscar Herrera. 2013. *"Efficient CPI-Based Taylor Rules in Small Open Economies."* Working Papers no. 694. Central Bank of Chile.

Ginker, Tim, and Tanya Suhoy. 2022. *"Nowcasting and Monitoring Real Economic Activity in Israel."* Bank of Israel Working Papers 2022.07. Bank of Israel.

Goldberg, Dror. 2024. *Gold, Fiat and Crypto*. Raanana.

O'Keeffe, Hannah, and Katerina Petrova. 2025. *"Component-Based Dynamic Factor Nowcast Model."* Federal Reserve Bank of New York Staff Reports, no. 1152, April.

Taylor, John B. 1993. *"Discretion versus Policy Rules in Practice."* Carnegie-Rochester Conference Series on Public Policy 39 (1): 195–214